

Shell Scripting Scenario(Self)

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Scenario 1: Evaluating if the file belongs to a user, if yes then give the file access to the user by checking if the file already exists in the directory.

Solution:

1. Check if the user provides the file name
2. Check if the file exists in the directory
3. Validate the ownership of the file
4. If the user is the actual owner of the file, then give read and write access

Shell Scripting

```
#!/bin/bash
```

```
FILE_PATH="$1"
```

```
CURRENT_USER=$(whoami)
```

```
# Check if file name is provided
```

```
if [ -z "$FILE_PATH" ]; then
```

```
    echo "Usage: ./secure_file_check.sh <file_path>"
```

```
    exit 1
```

```
fi
```

```
# Check if file exists
```

```
if [ ! -e "$FILE_PATH" ]; then
```

```
    echo "File does not exist."
```

```
    exit 2
```

```
fi
```

```

# Get file owner

FILE_OWNER=$(stat -c '%U' "$FILE_PATH")

# Check ownership

if [ "$FILE_OWNER" = "$CURRENT_USER" ]; then

    echo "File belongs to you."

    # Give read and write permission only to owner

    chmod 600 "$FILE_PATH"

    echo "Access granted. Permissions set to 600."

else

    echo "Access denied. File belongs to $FILE_OWNER."

fi

```

```

USTR+307420@DZTBKT3 MINGW64 ~
$ cd desktop

USTR+307420@DZTBKT3 MINGW64 ~/desktop
$ cd Practice

USTR+307420@DZTBKT3 MINGW64 ~/desktop/Practice
$ mkdir shell_practice

USTR+307420@DZTBKT3 MINGW64 ~/desktop/Practice
$ cd shell_practice

USTR+307420@DZTBKT3 MINGW64 ~/desktop/Practice/shell_practice
$ touch file_validation.sh

USTR+307420@DZTBKT3 MINGW64 ~/desktop/Practice/shell_practice
$ chmod +x file_validation.sh

USTR+307420@DZTBKT3 MINGW64 ~/desktop/Practice/shell_practice
$ touch myfile.txt

USTR+307420@DZTBKT3 MINGW64 ~/desktop/Practice/shell_practice
$ ls -l myfile.txt
-rw-r--r-- 1 USTR+307420 4096 0 Feb 18 10:48 myfile.txt

USTR+307420@DZTBKT3 MINGW64 ~/desktop/Practice/shell_practice
$ ./file_validation.sh myfile.txt
File belongs to you.
Access granted. Permissions set to 600.

USTR+307420@DZTBKT3 MINGW64 ~/desktop/Practice/shell_practice
$ ./file_validation.sh test.txt
File does not exist.

USTR+307420@DZTBKT3 MINGW64 ~/desktop/Practice/shell_practice
$ |

```